Ejection Fraction Estimation Using an R(2+1)D-18 Convoluted Neural Network

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Abstract

Left Ventricular Ejection Fraction (LVEF) is a key clinical metric used to assess cardiac function and predict conditions such as heart failure, but accurate estimation from echocardiographic data remains challenging due to variability in image quality and operator dependence. This paper presents an R(2+1)D-18 Convolutional Neural Network for estimating LVEF directly from ultrasound video sequences, leveraging decomposed spatiotemporal convolutions to capture both spatial and temporal features. The work replicates and adapts Stanford's "Video-Based AI for Beat-to-Beat Assessment of Cardiac Function", incorporating architectural modifications and Kinetics-400 pre-trained weights to improve feature learning. We additionally introduce an uncertainty-aware inference mechanism that enables the model to defer low-confidence predictions for human review, achieving >=95% accuracy on approximately 76% of cases while flagging the remaining 24%, primarily borderline or ambiguous cases, for clinician assessment. The model is further applied to classify patients into clinically relevant LVEF categories, supporting risk stratification and diagnostic decision-making, demonstrating its potential for safe and reliable clinical deployment.

1 Introduction

Accurate estimation of Left Ventricular Ejection Fraction (LVEF) is a key challenge in cardiac imaging due to variability in image quality and dependence on operator expertise (Ouyang et al., 2020). As LVEF is widely used in the diagnosis and monitoring of cardiovascular conditions such as heart failure (McDonagh et al., 2021), improving estimation consistency is essential for effective patient care.

Recent advances in video-based deep learning have demonstrated strong performance in cardiac function estimation. In this work, we replicate and extend Stanford's "Video-Based AI for Beat-to-Beat Assessment of Cardiac Function" using an R(2+1)D-18 convolutional neural network. Unlike standard 3D convolutional networks, the R(2+1)D architecture factorises 3D convolutions into separate spatial (2D) and temporal (1D) operations, enabling more efficient learning of spatiotemporal features. The model is initialised with Kinetics-400 pre-trained weights to leverage prior knowledge of motion and temporal patterns in video data.

Beyond predictive performance, a key contribution of this work is the integration of uncertainty-aware decision-making. Rather than forcing predictions on all inputs, the model identifies low-confidence cases and defers them for expert review. This approach improves overall reliability and aligns with real-world clinical workflows, where uncertain cases are routinely escalated to specialists. The key contributions of this work are: (i) replication of a state-of-the-art video-based LVEF estimation framework; (ii) incorporation of uncertainty-aware inference; and (iii) application to both regression and classification of clinically relevant LVEF categories.

2 Related Work

Echocardiographic estimation of LVEF has traditionally relied on manual measurement and geometric assumptions derived from 2D imaging (Litjens et al., 2017). While clinically established, these methods depend heavily on operator expertise and image quality, limiting reproducibility and scalability.

Recent advances in deep learning have enabled end-to-end estimation of LVEF directly from echocardiographic data. Ouyang et al. (2020) introduced a video-based framework that integrates spatial and temporal information from ultrasound sequences using convolutional neural networks, showing that modelling temporal dynamics improves performance over frame-based approaches using static snapshots. However, this approach focuses on point estimates of LVEF and does not explicitly model prediction uncertainty, limiting robustness in clinically ambiguous cases.

To better capture spatiotemporal structure, architectures such as R(2+1)D decompose 3D convolutions into separate spatial and temporal components. Compared to standard 3D CNNs, this factorisation improves optimisation stability and data efficiency, making it well suited to medical video datasets. Despite these advantages, prior echocardiography applications have mainly focused on accuracy optimisation rather than decision reliability or clinical safety.

More broadly, uncertainty-aware deep learning has been explored to improve trustworthiness in medical AI systems (Geifman and El-Yaniv, 2017). In safety-critical domains such as cardiology, the ability to defer uncertain predictions is particularly important. Alternative modalities such as ECG-based deep learning offer scalable screening but lack direct spatiotemporal representation of cardiac mechanics, making them complementary rather than substitutes for imaging-based techniques (Chen et al., 2022).

Building on these limitations, this work replicates the framework of Ouyang et al. (2020) using an R(2+1)D-18 backbone for improved spatiotemporal feature learning, while introducing uncertainty-aware inference that allows the model to defer low-confidence predictions for clinician review.

3 Data

The dataset used is EchoNet-Dynamic, released as part of Stanford University's study on video-based assessment of cardiac function. It consists of over 10,000 echocardiogram (ultrasound) videos from real patients, capturing motion of the left ventricle. Each sample is a sequence of frames with an associated ground-truth LVEF value used as the regression label, making the dataset well-suited to spatiotemporal models such as 3D CNNs. While large for a medical task, the dataset remains limited compared to general computer-vision corpora and exhibits high variability in heart positioning, ultrasound noise, and operator technique, which makes the problem realistically challenging.

3.1 Pre-processing

Metadata was loaded from the dataset file list and used to apply the official train/validation/test split, with file checks ensuring all videos and labels were present. Each sample consisted of 32 grayscale frames at 112x112 resolution. Frames were extracted with a temporal stride of 2 from a contiguous 64-frame window, matching the EchoNet-Dynamic configuration and covering approximately one cardiac cycle. Frames were resized to 112x112, scaled to [0, 1], and standardised using mean and standard deviation values computed from 100 randomly sampled training videos. Fixed-length clips were sampled using a random start position during training and a centred clip during validation and testing; shorter videos were evenly sampled and zero-padded. LVEF labels were retained as continuous values for regression and grouped into three clinically relevant classes for classification: Reduced (<40%), Mildly Reduced (40–49%), and Preserved (>=50%). Training augmentation included 12-pixel padding, random 112x112 cropping, and random horizontal flipping (p=0.5), applied consistently across all frames to preserve temporal coherence. To address class imbalance toward Preserved EF, inverse-frequency class weights were computed from the training split and applied in the cross-entropy loss.

4 Methods

The proposed system estimates LVEF directly from echocardiographic video using a single network with two task-specific output heads. A shared R(2+1)D-18 backbone produces a spatiotemporal embedding from a 32-frame grayscale clip, which is passed through a small shared multilayer perceptron (MLP) and then split into a regression head predicting continuous EF in [0, 1] and a classification head predicting one of three clinically relevant EF categories (Reduced, Mildly Reduced, Preserved). Both heads are optimised jointly with a weighted sum of mean-squared-error and cross-entropy losses, allowing the regression task to anchor the embedding in continuous EF space while the auxiliary classification head encourages separation of clinically meaningful categories.

4.1 Backbone: R(2+1)D-18

We use the R(2+1)D-18 architecture of Tran et al. (2018), which factorises each 3D convolution of size kt x kh x kw into a spatial convolution (1 x kh x kw) followed by a temporal convolution (kt x 1 x 1). Compared with full 3D convolutions, this decomposition doubles the number of non-linearities along the spatiotemporal path without increasing parameter count, decouples spatial and temporal optimisation (improving convergence on smaller medical-video datasets), and matches the action-recognition prior captured by Kinetics-400 pre-training.

We initialise the backbone with the R2Plus1D_18_Weights.KINETICS400_V1 weights from torchvision. Because echocardiograms are single-channel, the first 3D convolution is replaced with a 1-channel variant whose weights are obtained by averaging the three RGB input slices of the pre-trained kernel along the channel axis, preserving the Kinetics-400 prior while accepting grayscale input directly.

4.2 Two-head design with a shared embedding

Stanford's EchoNet-Dynamic uses a single regression head. We extend this by replacing the original 400-class Kinetics classifier with a shared MLP and two parallel linear heads. The 512-dim post-pooling backbone feature is projected through Linear (512 → 256) → ReLU → Dropout (0.5) → Linear(256 → 128) → ReLU to produce a 128-dim shared embedding, which feeds a regression head Linear (128 → 1) and a classification head Linear (128 → 3). Sharing the embedding lets the classification signal regularise the regression representation (and vice versa) without the cost of two independent networks or the asymmetric coupling of a cascaded design. A "no shared head" variant is retained as an ablation (Section 5.6).

4.3 Joint loss

The two heads are trained jointly with

$$L = \lambda reg\, MSE(\hat{y}\, reg\,,\, y) + \lambda cls\, CE(\hat{y}\, cls\,,\, c)$$

where y is the continuous EF label, c is its derived category, and the loss weights are lambda_reg = 1.0 and lambda_cls = 0.1. The classification weight is deliberately small: classification is treated as an auxiliary regulariser rather than a competing primary objective. Cross-entropy uses inverse-frequency class weights computed from the training split, since EchoNet-Dynamic is heavily skewed toward Preserved EF.

4.4 Optimisation

Following Ouyang et al. (2020), we train with SGD (momentum 0.9), learning rate 1e-4, StepLR decaying by 0.1 every 15 epochs, batch size 16, and up to 45 epochs with early stopping on validation loss (patience = 10). The best-validation-loss snapshot is saved as best_r2plus1d.pt. SGD is retained rather than switching to Adam to stay close to the original protocol and enable a direct comparison with Stanford's reported numbers.

4.5 Inference and decision pipeline

At inference, the trained model produces both a continuous EF estimate and a 3-way categorical distribution from a single forward pass. A post-hoc evaluation pipeline is then applied without modifying model weights:

1. Test-time augmentation (TTA): 10 random 32-frame clips are sampled per video, with regression outputs and classification logits averaged.
2. Temperature scaling: a single scalar T is learned on validation logits to reduce softmax overconfidence.
3. Threshold optimisation: default 40% / 50% category boundaries are replaced with thresholds (T1, T2) selected by grid search on validation regression outputs. Seven strategies were compared, including Baseline, Macro-F1, Recall-optimised, Ensemble, Bayesian cost-minimisation, and a Balanced strategy maximising Mildly Reduced recall while maintaining accuracy ≥80%.

The Balanced strategy with T1 = 46% and T2 = 56% was selected. Final categorical predictions were obtained by re-thresholding the TTA-averaged regression output using the Balanced thresholds. The temperature-scaled classifier softmax was used as the confidence signal for the uncertainty-aware deferral rule in Section 5.4, where predictions below a confidence threshold t were deferred for clinician review.

4.6 Implementation

The model is implemented in PyTorch using torchvision.models.video.r2plus1d_18 and trained on a single NVIDIA RTX 5090 GPU. All hyperparameters and dataset paths are centralised in src/config.py, and the full pipeline (data loading, training, evaluation, Grad-CAM, ablations, calibration) is orchestrated from main.py.

5 Experiments

All experiments used the official EchoNet-Dynamic splits, with the model trained as described in Section 4.6. At test time we apply the post-hoc clinical evaluation pipeline of 4.7: ten-clip test-time augmentation per video, temperature scaling of the classification logits (T = 0.25, learned on the validation set), and recall-aware re-thresholding of the regression output (Balanced strategy, T1 = 46%, T2 = 56%). For uncertainty-aware inference, confidence is the maximum temperature-scaled softmax probability, and the deferral threshold is determined by sweeping over the test set within the uncertainty analysis routine.

5.1 Evaluation Metrics

For regression, we report Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). For classification, we report accuracy, per-class precision, recall, F1-score, ROC-AUC (one-vs-rest), and the confusion matrix. Calibration of the classification head is summarised by the Expected Calibration Error (ECE) computed over 10 equal-width confidence bins. To evaluate uncertainty-aware inference, we report coverage (the proportion of predictions retained after thresholding) and accuracy at coverage (performance on the retained subset).

5.2 Baseline and Model Performance

For LVEF regression, the model achieved a Mean Absolute Error of 4.10% (95% CI 3.90–4.30%), a Root Mean Squared Error of 5.55% (5.21–5.88%), and a coefficient of determination $R^2 = 0.794$ (0.7634–0.8204) on the test set. These results closely match the Stanford study (MAE 4.1%, RMSE 5.3%) (Ouyang et al., 2020). Figure 1 shows predicted vs. actual EF, illustrating strong correlation and low dispersion around the ideal line. Training curves (Figure 4) show training and validation loss decreasing steadily and converging closely, indicating limited overfitting; MAE stabilises near 4% and the auxiliary classification head's training-time accuracy (under the default 40/50 thresholds) converges at approximately 87–89%.

A Bland–Altman analysis (Figure 2) shows a small bias of +0.41% with most samples within the 95% limits of agreement (−10.43% to +11.26%), indicating good agreement between predicted and ground-truth EF.

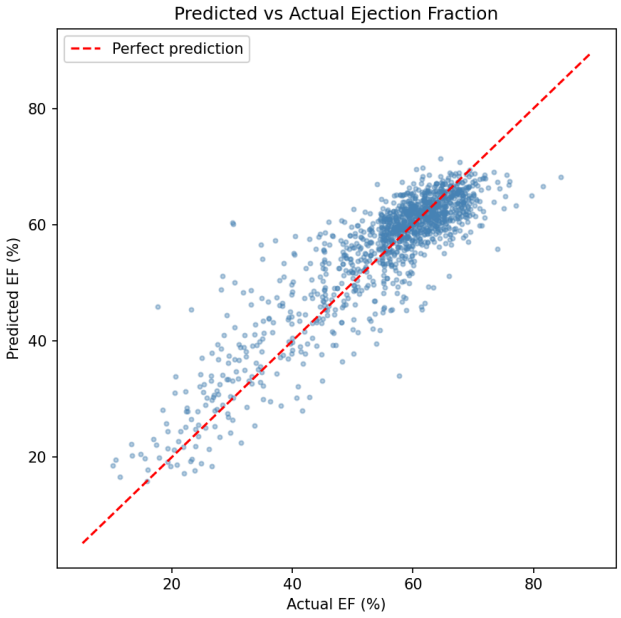


Figure 1: Predicted vs Actual EF scatter graph

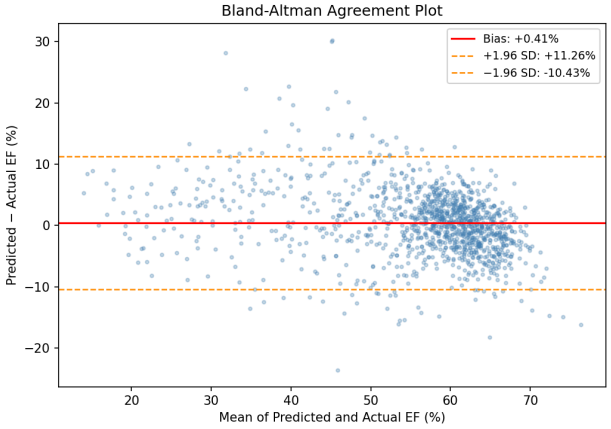


Figure 2: Bland–Altman agreement plot

For categorical classification under the post-hoc Balanced (46/56) strategy, the model achieved an overall accuracy of 80.6%, ROC-AUC of 0.932 (one-vs-rest), and an Expected Calibration Error of 0.035 after temperature scaling. Per-class metrics are summarised in Table 1, and the confusion matrix in Figure 3 shows that most misclassifications occur between adjacent categories (e.g. Mildly Reduced vs Preserved), as expected given the continuous nature of LVEF. The Balanced strategy also reduces dangerous Reduced → Mildly-Reduced mis-triages from 33 patients (rate 0.206) under the default 40/50 thresholds to 20 patients (rate 0.125), while raising Mildly-Reduced recall from 0.392 to 0.560.

Class	Precision	Recall	F1-score
Reduced (<40%)	0.72	0.85	0.78
Mildly Reduced (40-49%)	0.29	0.56	0.38
Preserved (≥50%)	0.97	0.83	0.90
Macro Avg	0.66	0.75	0.69
Weighted Avg	0.88	0.81	0.83

Table 1: Classification metrics

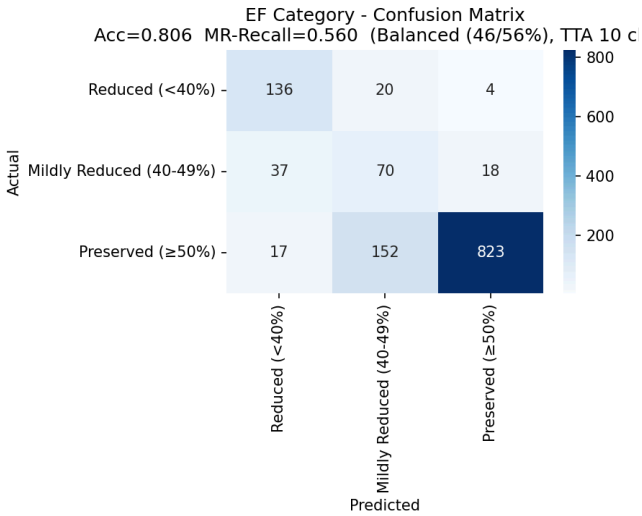


Figure 3: Confusion matrix

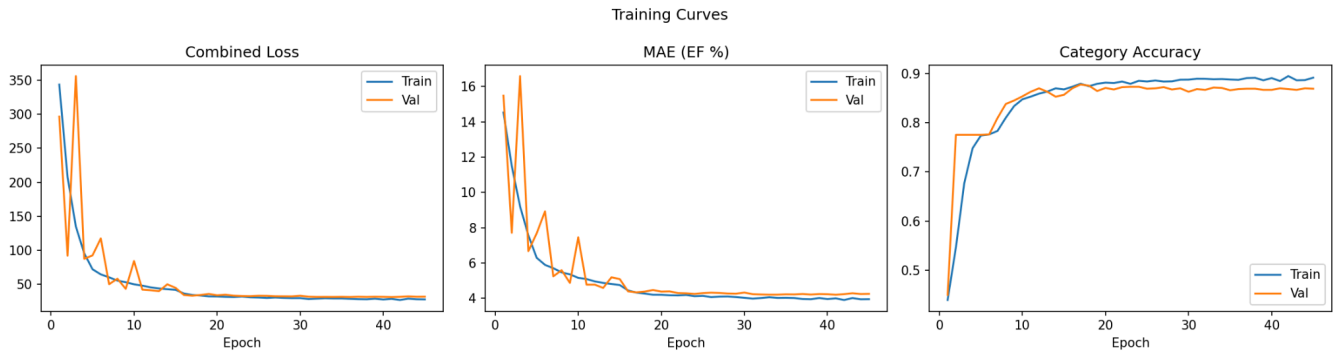


Figure 4: Training curves showing combined loss, MAE, and classification accuracy across training epochs.

5.3 Uncertainty-Aware Inference Results

To improve reliability, a confidence-based deferral mechanism was applied on top of the Balanced strategy. Confidence is taken to be the maximum temperature-scaled softmax probability; predictions below a threshold t are deferred for review while the retained predictions use the Balanced regression-derived class label.

The mechanism produces a clear coverage–accuracy trade-off. At t = 0.86 the model achieves 95% accuracy on 75.7% of predictions, with the remaining 24.3% deferred for clinician review. Figure 5 illustrates the coverage–accuracy relationship and confirms the model's ability to rank predictions by confidence.

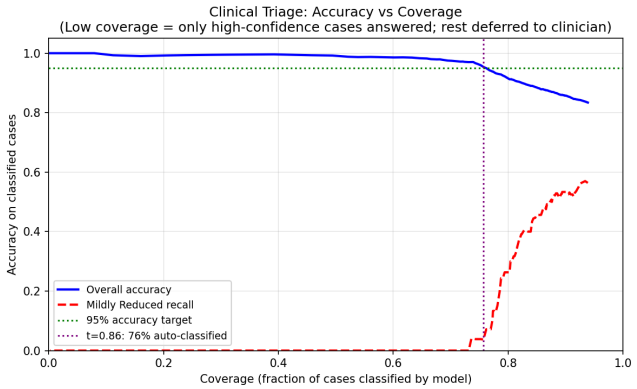


Figure 5: Accuracy vs Coverage Plot

5.4 Qualitative Analysis

High-confidence correct predictions were typically associated with clear cardiac motion and high-quality imaging, where ventricular boundaries were well-defined across the cycle. Misclassifications were concentrated near class boundaries (e.g. 40-49% vs >=50%), where small numerical differences produce different labels, errors that are typically clinically minor. Deferred (low-confidence) cases consistently exhibited poor image quality, ambiguous or irregular motion, challenging probe positioning, or borderline EF values, indicating that the model appropriately recognises uncertainty in cases where even human interpretation is difficult.

5.5 Ablation Study

All ablations are evaluated under the same baseline 40/50 thresholds, without test-time augmentation or temperature scaling, so that comparisons across configurations are like-for-like. Under this protocol the proposed model attains a test accuracy of 0.871 and an MAE of 4.10%; the post-hoc Balanced strategy reported in 5.3 trades approximately 6.5 percentage points of raw accuracy for a 17-point gain in Mildly-Reduced recall and a 40% reduction in Reduced to Mildly-Reduced mis-triage.

Table 2 reports the contribution of key design choices, with each experiment isolating a single component while keeping all other settings fixed.

Configuration	MAE (%)	Accuracy
32-frame baseline	4.10	0.8708
16-frame baseline	4.62	0.8463
No pre-training	10.24	0.8090
Regression-only	4.38	0.7756
No shared head	4.19	0.8439

Table 2: Ablation study results

The 16-frame baseline produced a higher MAE and lower classification accuracy than the proposed 32-frame configuration, suggesting that reduced temporal context limits the model's ability to capture motion across the complete cardiac cycle. Removing Kinetics-400 pre-training caused a substantial regression degradation (MAE 4.10% → 10.24%), confirming that transfer learning from large-scale video data provides a strong spatiotemporal prior, particularly important for precise continuous EF estimation. The regression-only configuration reduced both regression and classification performance, indicating that the auxiliary classification objective regularises the shared representation. The no-shared-head variant matched the proposed model on MAE but slightly exceeded it on accuracy, suggesting that the shared embedding primarily benefits regression consistency; the proposed design nevertheless gives the most balanced multi-task performance.

5.6 Grad-CAM Visualisation Analysis

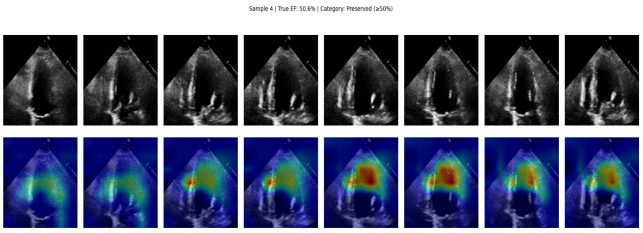


Figure 6: Gradient-weighted Class Activation Mapping (Grad-CAM) on a preserved patient

Grad-CAM was used to visualise the spatial regions most strongly influencing predictions. In high-confidence cases, attention concentrated around ventricular boundaries and myocardial motion during systole and diastole, consistent with the clinical procedure for estimating ejection fraction. In challenging or deferred cases, activation maps were noticeably more diffuse, particularly under poor ultrasound quality, shadowing, or ambiguous ventricular visibility. This supports the conclusion that the model bases predictions on anatomically meaningful regions rather than background artefacts, improving transparency and reinforcing the reliability of the uncertainty-aware framework.

5.7 Limitations and Failure Cases

EchoNet-Dynamic, while large for a medical dataset, represents a single domain and may limit generalisation across hospitals, devices, and patient populations

(Oakden-Rayner et al., 2019). Performance is reduced when ultrasound quality is poor or the left ventricle is partially occluded. Borderline EF values near clinical thresholds produce apparent misclassifications that may not be clinically significant. Confidence is derived from softmax probabilities, which are not true Bayesian uncertainty estimates and may be overconfident in some cases (Gal and Ghahramani, 2016), methods such as Monte Carlo dropout or deep ensembles could improve calibration. Fixed 32-frame clips may not fully represent longer or irregular cardiac dynamics. This work focuses on the EF regression and classification components of EchoNet-Dynamic; left ventricular segmentation, external multi-site validation, and prospective clinician comparison studies were beyond the scope of this replication. Finally, the system is not intended to replace clinicians; the uncertainty-aware framework is designed to support decision-making by automating high-confidence cases and deferring ambiguous ones for expert review.

5.8 Summary of Findings

The R(2+1)D-18 architecture effectively captures spatiotemporal cardiac motion for LVEF estimation, achieving regression and classification performance comparable to prior work. Kinetics-400 pre-training significantly improves accuracy, while uncertainty-aware inference enables 95% accuracy on roughly 76-77% of high-confidence predictions and defers ambiguous cases. The system is well-suited to a clinical decision-support setting where reliable automation is combined with clinician oversight.

6 Conclusion

This paper presented an R(2+1)D-18 CNN for LVEF estimation from echocardiographic video, achieving regression performance matching the Stanford EchoNet-Dynamic benchmark (MAE 4.1%, RMSE 5.55%, $R^2 = 0.794$). The dual-head architecture demonstrated that multi-task learning provides meaningful regularisation across regression and classification.

The primary contribution beyond replication is a post-hoc clinical evaluation pipeline combining TTA, temperature scaling, recall-aware re-thresholding, and uncertainty-aware deferral. The system achieves ≥95% accuracy on 76% of test cases at t = 0.86 while deferring the remaining 24%, predominantly borderline or poor-quality samples, for clinician review. The Balanced (46/56) thresholding strategy reduces Reduced → Mildly-Reduced mis-triages by 40% and raises Mildly-Reduced recall from 0.39 to 0.56 relative to default thresholds. Grad-CAM visualisations confirmed anatomically meaningful attention.

Transfer learning from large-scale video data is critical (MAE increases from 4.1% to 10.24% without it); temporal context improves precision; and softmax-based confidence provides an effective, lightweight selective-prediction signal. Future work includes better-calibrated uncertainty estimates, multi-site validation, and integration into a clinical decision-support interface.

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